

between two tracks or the total size of a card. You can just right click and drag to open a ruler in green for your help.

Keyboard shortcuts. From selecting multiple graphic elements to drawing a PCB pad, there are various drawing commands available in FidoCADJ. These allow you to rapidly select various features that let you design faster (as shown in the table), without having to manipulate your mouse.

Flexible graphics. It also offers three different zoom settings for graphical flexibility with the buttons like *Fit*, *Show Grid* and *Snap to Grid*. You can automatically select the most suitable zoom settings in order to show the whole drawing on the screen by pressing *Fit* button. *Show Grid* makes it easy to toggle between visible and invisible grids. Finally, the elements added will stick to the nearest step of the grid utilising the *Snap to Grid* button. (Note: To carefully align various elements, *Alt* key should be kept pressed while using the other cursor keys.)

Partial customisation of command bar. FidoCADJ also allows you to partially customise the command bar. In particular, you can either choose to see the icon on each button or both the icon and its text description. These two selectable formats for the icons can be changed from the menu *File* → *Options*. Any change in the settings will be applied at the restart of the application.

Easy search and navigation. The libraries include all the standard symbols used in electrical schematics and a wide selection of footprints for drawing PCBs. A quick search text field allows for the quick searching inside the included libraries and the results appear in the form of a tree, allowing you to easily navigate through by typing up and down arrow keys.

Macros. The FidoCADJ application offers you a quick access to all the electrical components, lets you preview them in a dedicated pane, and insert them into the working environment with the help of just a few clicks. The tool includes these electrical components in the form of various macros or symbols in the libraries. A tree list at the right side of the window enables easy

Installing FidoCADJ

Installation file can be found in EFY Plus DVD that came with this magazine. FidoCADJ is a Java-based tool, so Java is a pre-requisite to run this tool on any platform.

The application is distributed as an executable .jar file, which is a Java archive. For many operating systems, you can just double click on the file to run the application, provided that a recent version of Java is installed on the system. The minimum Java version needed to run FidoCADJ is the version 1.5, which has been around for a few years now.

But, in some cases it might be useful to run the application from a command line (for example, the terminal in the Unix systems, or the MS-DOS Prompt in Windows). To do so, you can use the following command lines:

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1. java -jar fidocadj.jar
2. java -jar fidocadj.jar ~/FidoCADJ/test.Fcd
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(To open a specific file with the application, include it in the command line)

Supported OS: Windows, Linux, MacOS (with Java)

HDD space required: Max 10MB

selection of any preset element in the drawing.

Export with effects. FidoCADJ allows you to export your drawings through several file formats, so you can create simple schematics for use. The generated electronic circuits can be printed or exported to PNG, JPG, SVG, EPS, PGF, PDF, or to a script file format. Also, you can apply anti-aliasing and black-and-white effects to the drawings. To export the current drawing, select *File* → *Export*. With *Black&White* option, you can print any visible layer in solid black.

For the bitmap file formats, it may be useful to enable the option *Anti aliasing*. While exporting to any vectorial format, it will be beneficial to specify the scaling factor. *Anti aliasing* reduces the annoying effect of the quantisation that is visible especially on diagonal lines.

Customised libraries. As mentioned above, FidoCADJ includes a collection of libraries of symbols and footprints (traditional and SMD) for assistance in your drawings. It allows you to customise and manage these by defining new symbols and libraries with the luxury of modifying the existing ones. In some specific cases you can also replace the existing libraries with the external ones.

An effective 2D design tool

Summarising, we can regard FidoCAD as an effective tool for designing elec-

Keyboard Shortcuts

Keyboard shortcuts	Command	Description
A or Spacebar	Select	For selection of different elements
R	Rotate	To rotate selected elements
S	Mirror	To mirror the selected elements
L	Line	For inserting a line
T	Text	For inserting a text-string
B	Bezier	For drawing a Bezier curve
P	Polyline	For drawing a polyline
K	Curve	For drawing an open or closed natural cubic spline curve
E	Ellipse	For drawing an ellipse
G	Rectangle	For drawing a rectangle
C	Junction	To insert an electrical junction
I	PCB Track	To draw a PCB track
Z	PCB Pad	To draw a PCB pad
Esc	Terminate	To terminate new insertions

tronics. Its intuitiveness and handy set of features make it an easy-to-use and learning tool for hobbyists. Moreover, being free, Java-based and multi-platform, it can be of great use.

But it need not be considered uniquely for electronics design as customisation of its libraries can prepare it for any type of 2D drawing and for many other situations, provided those specific libraries can be made available.

FidoCADJ is the official drawing tool of ElectroYou, an Italian social network for electronics. It is a free GPL Licensed software tool that can be downloaded for free at <http://fidocadj.sourceforge.net/>. You can participate in the forums and blogs related to this tool at <http://sourceforge.net/p/fidocadj/discussion/997486> ●

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